

EXECUTIVE SUMMARY

- To build a machine learning system capable of predicting the operational status of waterpoints across Uganda and generating actionable, population-weighted priority scores for maintenance intervention.
- Starting from a raw global dataset of 832,638 waterpoints, the data was rigorously engineered into a clean dataset of 130,842 records.
- The model successfully identified that 40,286 waterpoints (30.8%) are at High Risk of failure, and a further 82,045 (62.7%) require immediate maintenance.

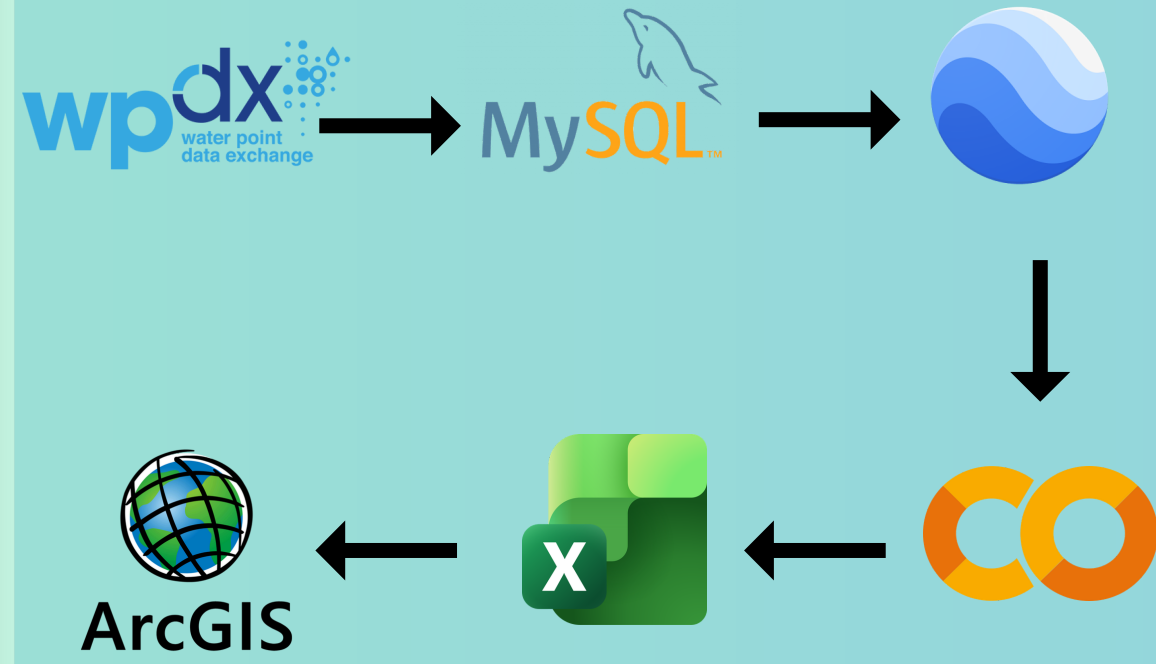
BACKGROUND

- In Uganda, a country of approximately 46+ million people, point-source facilities like boreholes and wells are the primary source of drinking water.
- Traditionally, waterpoint failure is identified reactively, forcing communities to wait months for repairs and leading to inequitable resource allocation.
- By shifting from a reactive to a predictive machine learning paradigm, government planners and NGOs can systematically identify and repair facilities before they fail.

PRIMARY GOAL

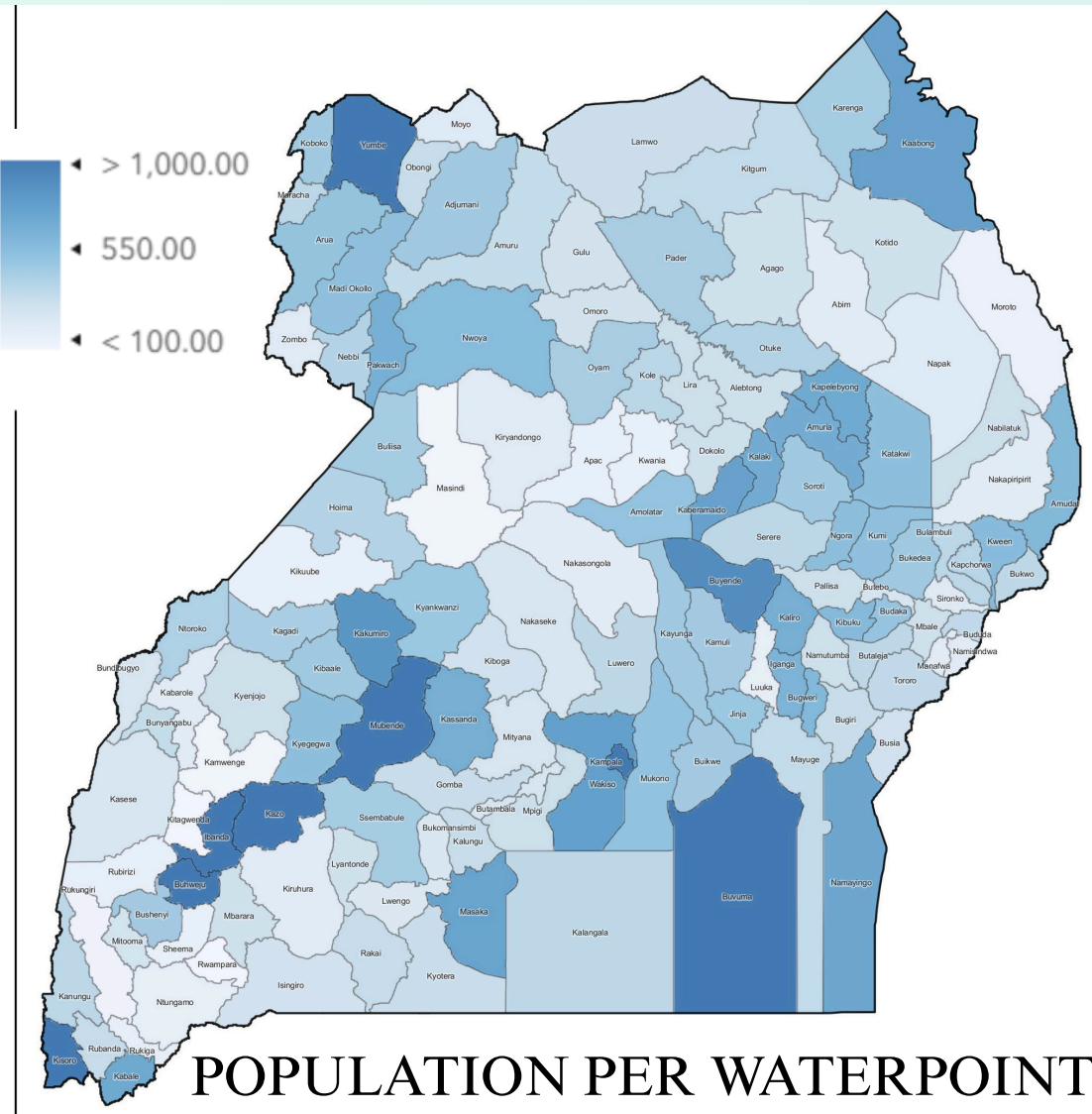
How can data-driven insights prioritize national infrastructure replacement and emergency interventions to resolve Uganda’s systemic water crisis?

PIPEPLINE



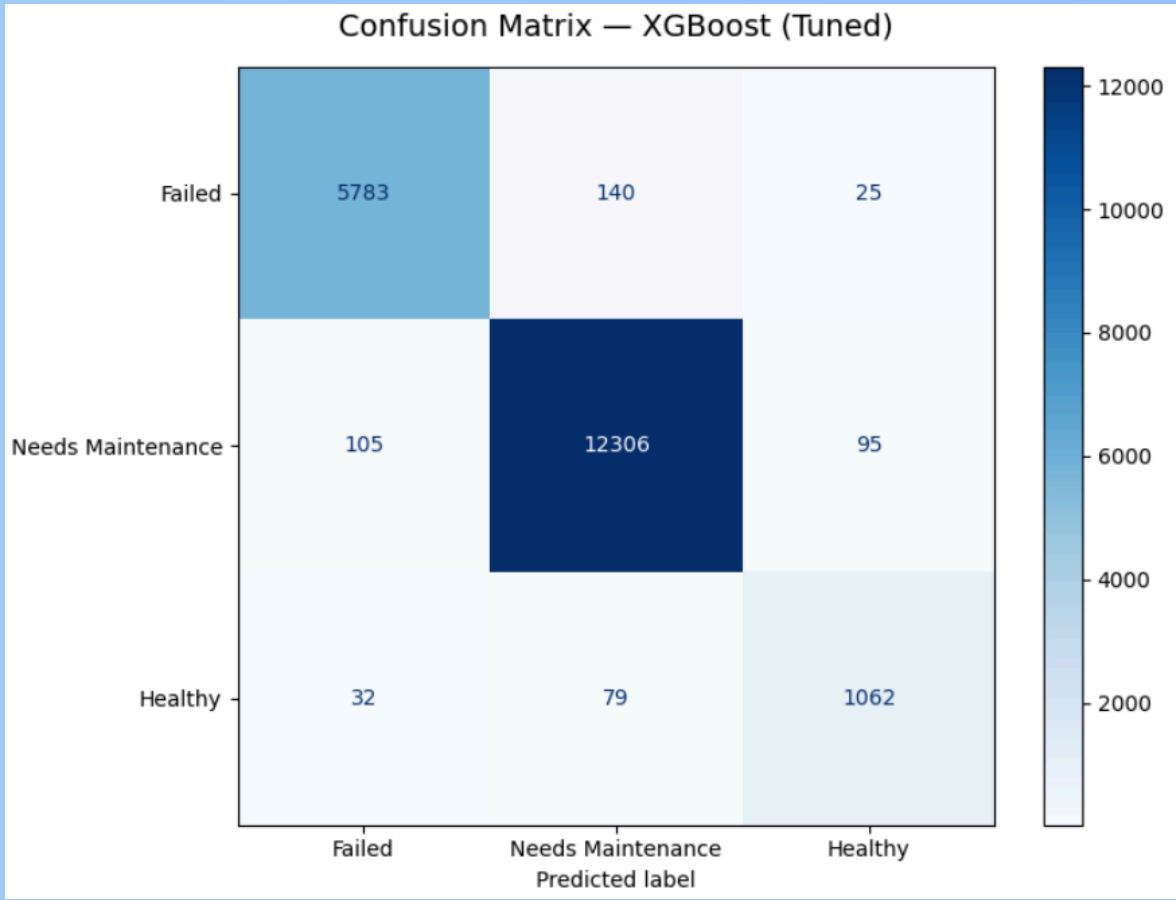
EXPERIMENTAL SETUP

- Data Engineering: Executed comprehensive data cleaning in MySQL Workbench, resolving missing installation years using a four-level hierarchical geographic imputation.
- Validation: Geographic coordinates were strictly validated using Google Earth KML polygons to ensure all records fell within Uganda's boundaries.
- Integration: Merged the Water Point Data Exchange (WPdx) records with district-level population datasets to engineer features capturing system stress.



EXPERIMENTAL TEST PLAN

- The 130,842 validated records were divided into a Training Set (70%), Validation Set (15%), and Test Set (15%).
- Stratified splitting was applied across all partitions to preserve the severe class imbalance of the target variable.
- The operational status of waterpoints was engineered into three predictive classes: Healthy, Needs Maintenance, and Failed.
- Multiple classifiers (Logistic Regression, Random Forest, Gradient Boosting) were evaluated before selecting XGBoost for hyperparameter tuning via Randomized SearchCV.



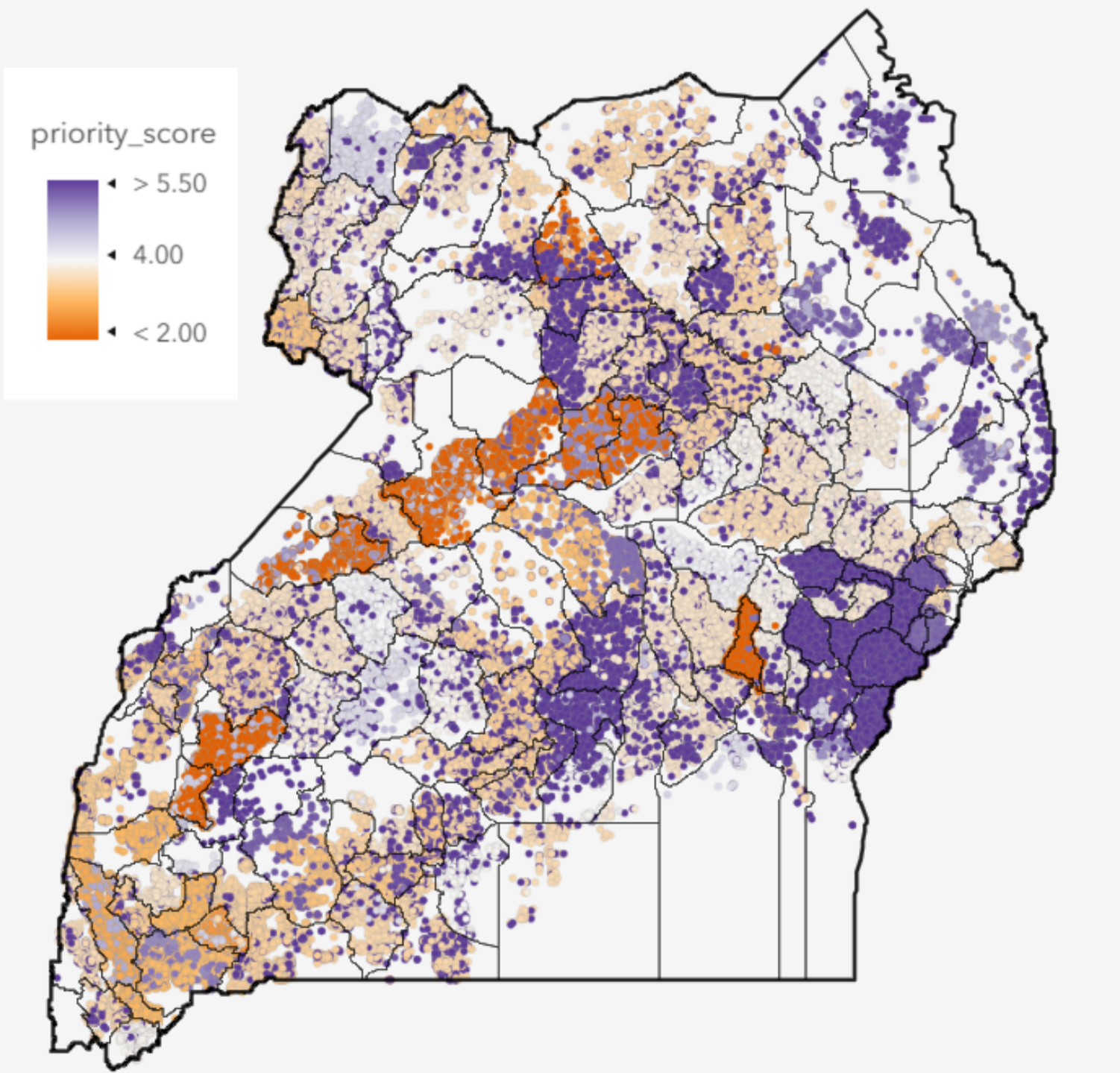
PRIORITY SCORE

Priority Mechanism:

Resource allocation cannot be based on mechanical risk alone. The map is driven by the project's composite Priority Score, which weighs the XGBoost risk score probability against the population burden of the surrounding district

CALCULATIONS

$$\text{priority_score} = \text{risk_score} \times \log(1 + \text{population_per_waterpoint})$$



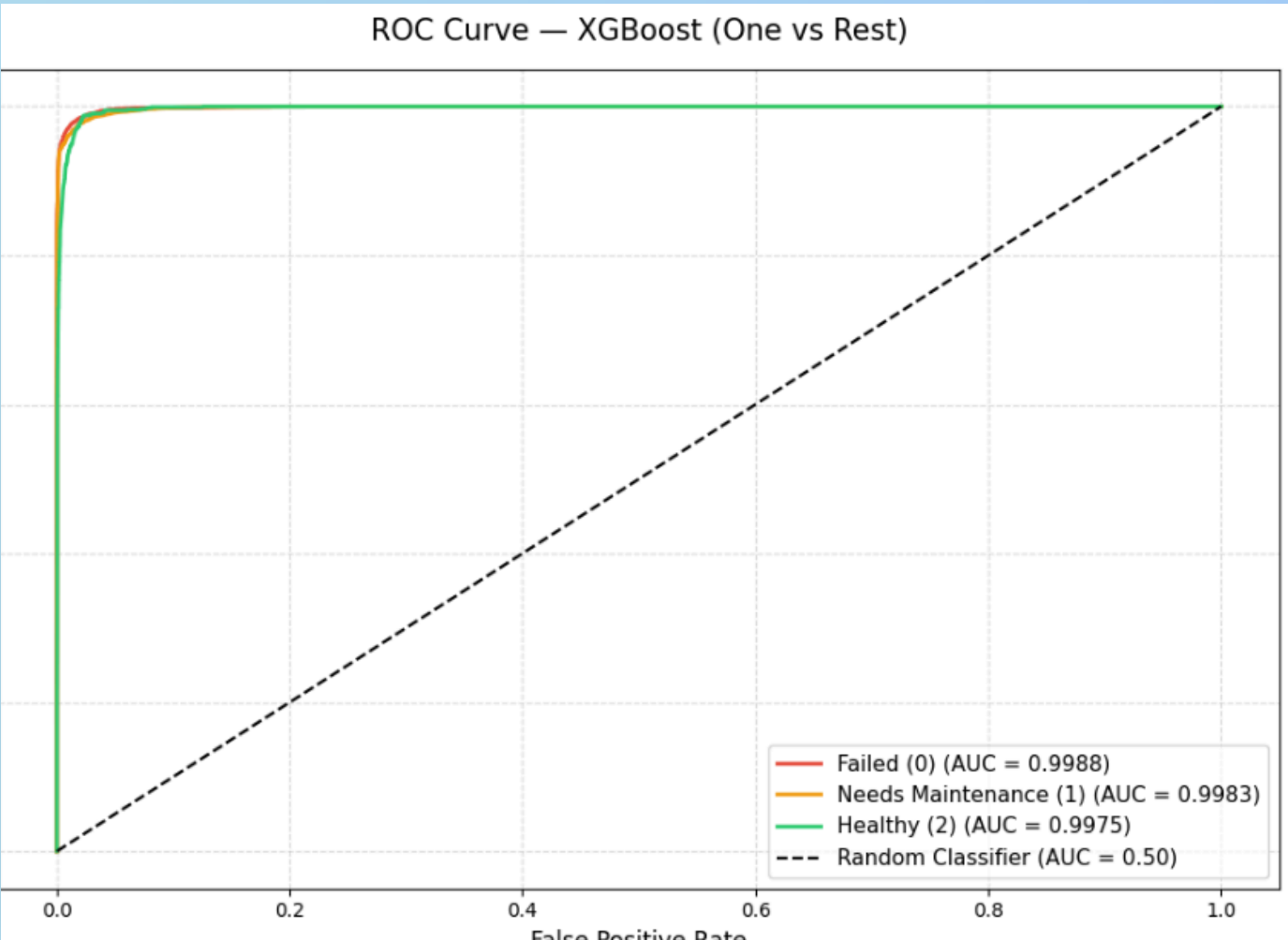
EXPERIMENTAL RESULTS

- The tuned XGBoost model achieved strong predictive performance, effectively distinguishing all three classes with the highest precision and recall on the dominant "Needs Maintenance" category.
- Feature importance analysis revealed that presence of water is the strongest predictor of failure, followed by management type and age groups.
- When the model was applied to all 130,842 Ugandan waterpoints, it classified 40,286 facilities (30.8%) as High Risk and 82,045 facilities (62.7%) as Medium Risk.

CLASSIFICATION REPORT

	PRECISION	RECALL	F1 SCORE	SUPPORT
ACCURACY			0.98	19627
MACRO AVG	0.95	0.95	0.95	19627
WEIGHTED AVG	0.98	0.98	0.98	19627

ROC CURVE



CONCLUSION

With only 6% of Uganda's surveyed waterpoints fully healthy , our XGBoost pipeline identified 40,286 high-risk facilities. To optimize resource allocation, we engineered a Priority Score weighting this mechanical risk against local population burdens. This enables policymakers to shift to proactive maintenance, targeting emergency interventions to the highest impact regions

ANALYSIS



PREDICTION

